# MyEyes

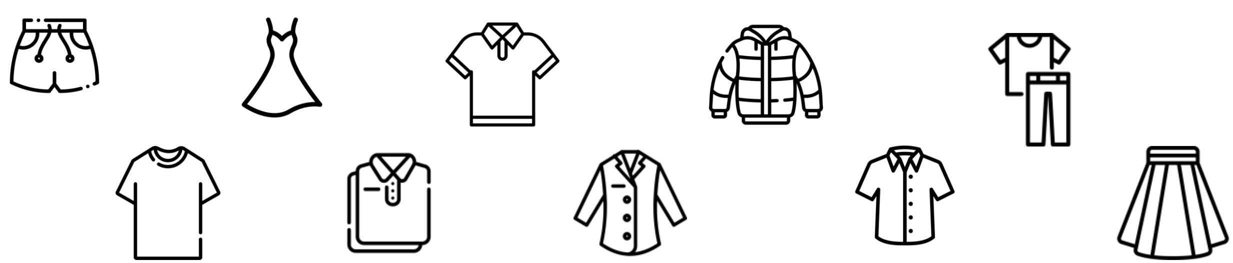
## *Garments Detection and Classification using Region based - CNNs*

Mariana Carvalho
Daniel Rocha
Vítor Carvalho

# Summary

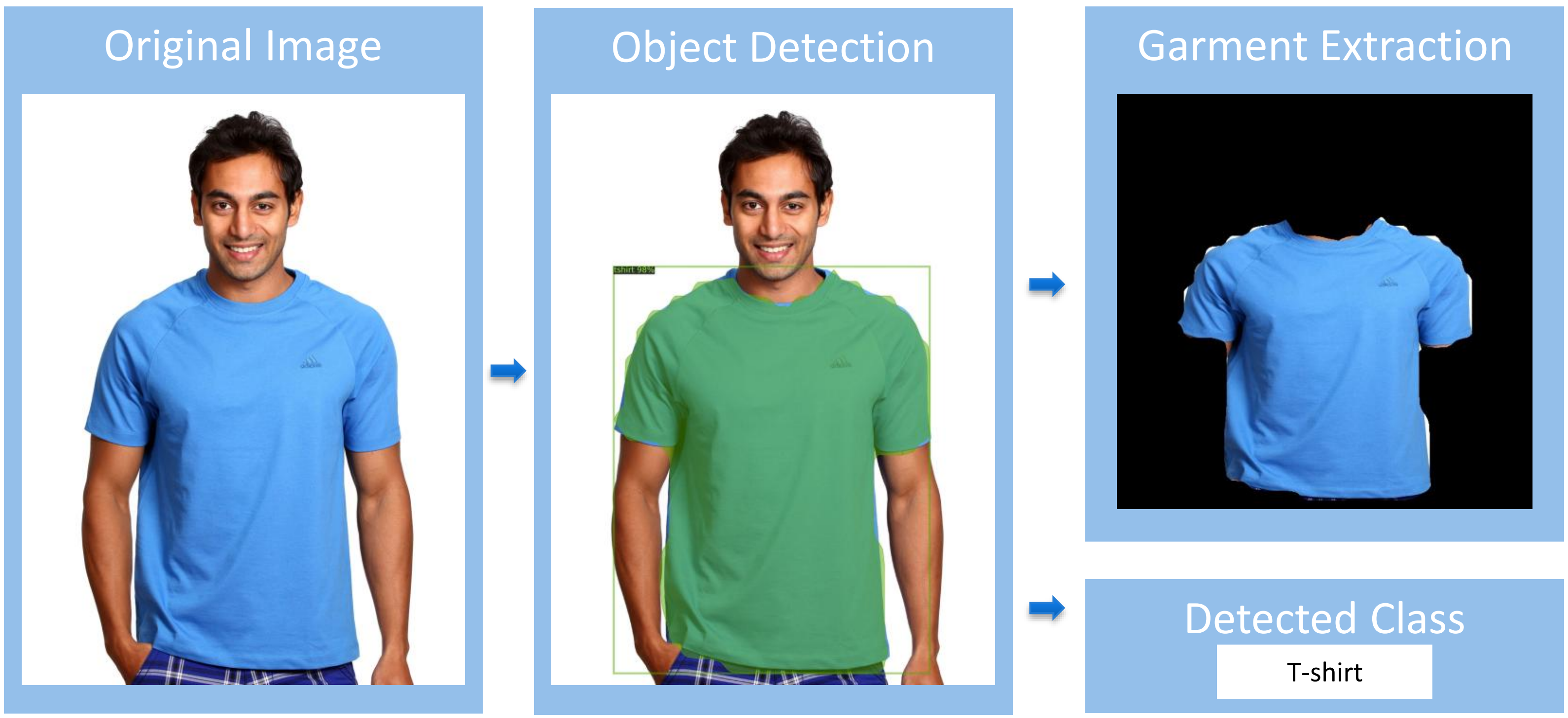
## 1. Motivation

Blind → Well-being → clothes



## 2. Objectives

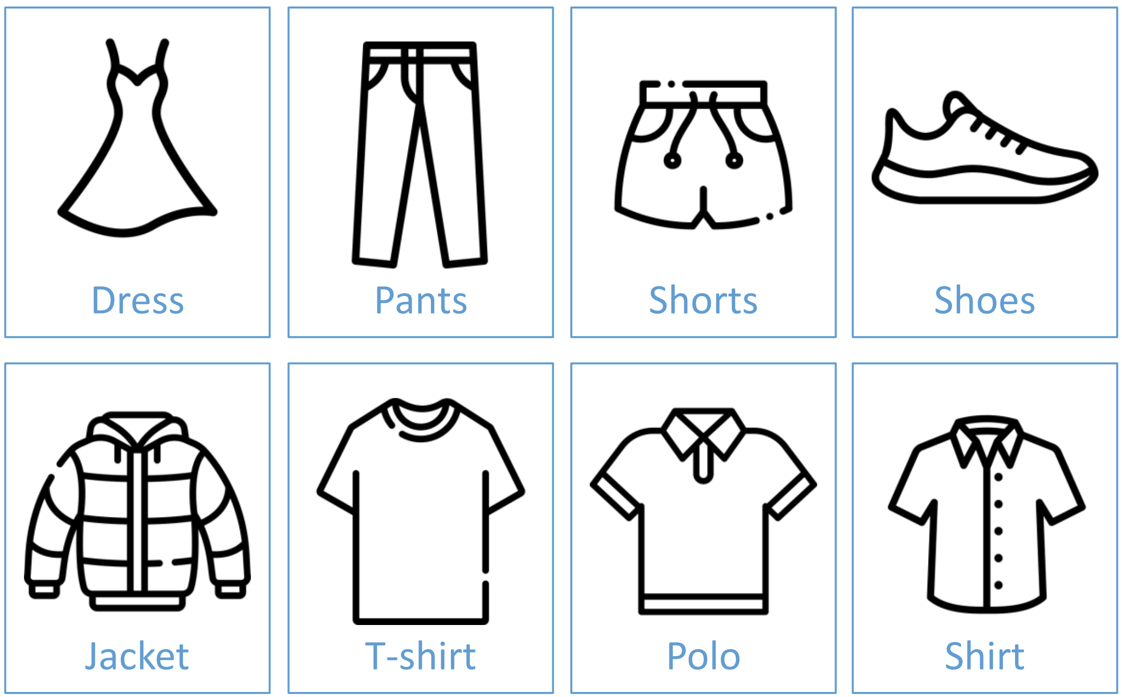
The main objective of this project is the automatic identification of clothing items using artificial intelligence algorithms.



## 3. Garments Identification and Classification

### 3.1. Dataset

Using polygonum labels, we developed dataset with a total of 1120 images distributed amongst 8 categories.



# 3. Garments Identification and Classification

## 3.2. Region Based - CNNs

Applying Region Based-CNNs with the help of mask R-CNN, we were able to train the dataset.

We tested under different condition of number of training iterations and different types of data augmentation in order to obtain the best possible results.

## 4. Data Cataloging

| | |
|---|---|
| Filename | 2242.jpg |
| Original image File Path | C:\Users\maria\detectron2\static\uploads\2242.jpg |
| Garment Class | T-shirt |
| Bounding Box | (array([ 701, 701, 701, ..., 2352, 2352, 2352], dtype=int64), array([709, 710, 711, ..., 355, 356, 817], dtype=int64)) |

# 5. Results (1)

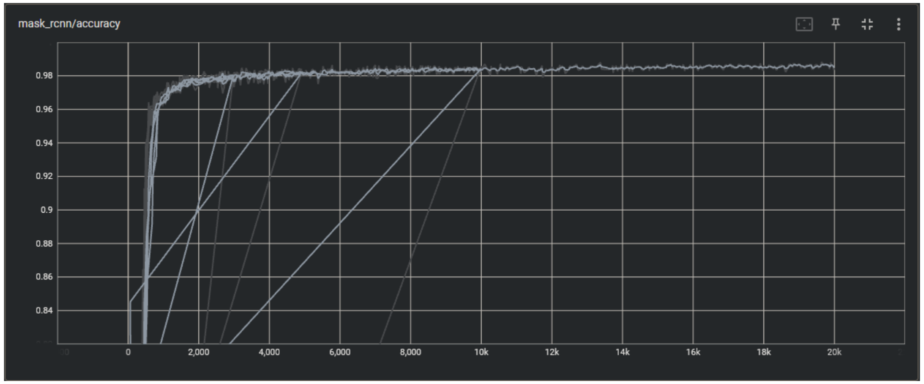
For 20 000 training iterations, we obtained the following results:

| CATEGORIES | TOTAL | DRESS | JACKET | PANTS | POLO | SHIRT | SHOES | SHORT | T-SHIRT |
|---|---|---|---|---|---|---|---|---|---|
| AP (%) | 76.53 | 80.1 | 70.26 | 78.89 | 68.75 | 65.74 | 97.93 | 81.08 | 69.49 |

Training was done with augmentation: flip(horizontal and vertical) and blur

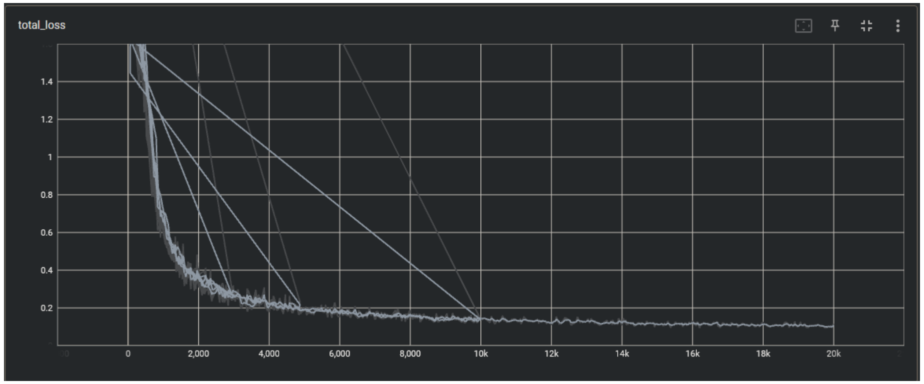
# 5. Results (2)

## Training Graphics




Accuracy

Loss



## 6. Final Remarks

In the future we are:

- augmenting the number of categories
- using images with different backgrounds
- expanding algorithms in an autonomous way

# Thank You!

Mariana Carvalho
Daniel Rocha
Vítor Carvalho



CENTRO 2020 PORTUGAL 2020 FCT Fundação para a Ciência e a Tecnologia UNIÃO EUROPEIA Fundo Europeu de Desenvolvimento Regional

TEchMA2023